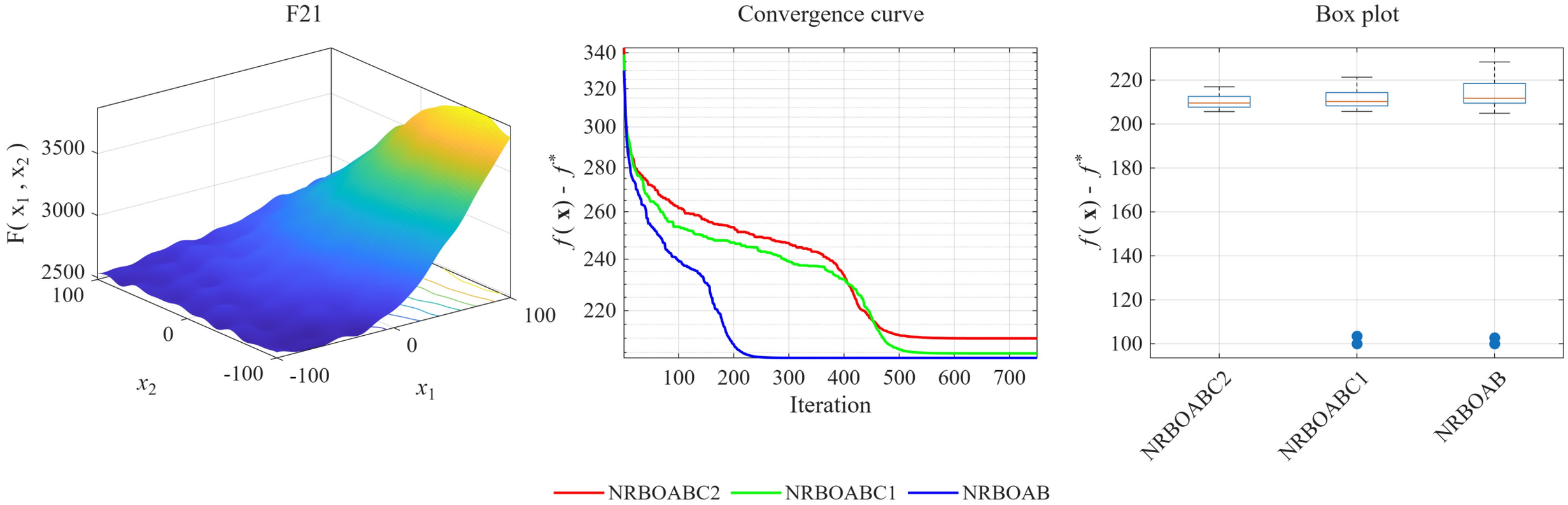

F21
F(x1 , x2)
3500
3000
2500
100
0
-100
x2
-100
0
100
x1
Convergence curve
f(x) - f*
340
320
300
280
260
240
220
100
200
300
400
500
600
700
Iteration
Box plot
f(x) - f*
220
200
180
160
140
120
100
NRBOABC2
NRBOABC1
NRBOAB
NRBOABC2
NRBOABC1
NRBOAB